



Entergy Operations, Inc.
River Bend Station
5485 U.S. Highway 61N
St. Francisville, LA 70775
Tel 225-381-4374

William F. Maguire
Site Vice President

RBG-47841

March 15, 2018

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555

Subject: Licensee Event Report 50-458 / 2017-004-01
River Bend Station – Unit 1
Docket No. 50-458
License No. NPF-47

RBF1-18-0047

Dear Sir or Madam:

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report. This document is a revision to Licensee Event Report 50-458 / 2017-004-00 dated May 22, 2017 and contains no commitments. If you have any questions, please contact Mr. Tim Schenk at 225-381-4177.

Sincerely,

A handwritten signature in black ink, appearing to read "W. F. Maguire", with a large, stylized loop at the end.

WFM / djp

Enclosure

cc: U. S. Nuclear Regulatory Commission
Region IV
1600 East Lamar Blvd.
Arlington, TX 76011-4511

NRC Sr. Resident Inspector
P. O. Box 1050
St. Francisville, LA 70775

INPO
(via ICES reporting)

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Central Records Clerk
Public Utility Commission of Texas
1701 N. Congress Ave.
Austin, TX 78711-3326

Department of Environmental Quality
Office of Environmental Compliance
Radiological Emergency Planning and Response Section
Ji Young Wiley
P.O. Box 4312
Baton Rouge, LA 70821-4312



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME River Bend Station – Unit 1	2. DOCKET NUMBER 05000-458	3. PAGE 1 OF 3
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4. TITLE
Loss of High Pressure Core Spray Safety Function During Surveillance Due to Malfunction of Test Return Valve

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
03	23	2017	2017	004	01	03	15	2018	FACILITY NAME	DOCKET NUMBER
										05000
									FACILITY NAME	DOCKET NUMBER
										05000

9. OPERATING MODE 1	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)			
	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
10. POWER LEVEL 100	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT

Tim Schenk, Manager – Regulatory Assurance

TELEPHONE NUMBER (Include Area Code)

225-381-4177

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX
(see text)									

14. SUPPLEMENTAL REPORT EXPECTED

☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE)☒ NO

15. EXPECTED SUBMISSION DATE

MONTH	DAY	YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On March 23, 2017, at 0028 CDT, with the plant operating at 100 percent power, the high pressure core spray system (HPCS) was declared inoperable due to a malfunction of a motor-operated valve (MOV) in the system. During a scheduled test, the HPCS pump test return valve to the suppression pool was given a "close" signal after having been opened for the test. The valve position lights indicated that it fully closed, but system flow parameters did not respond as expected. An operator went to the valve and reported that it appeared that the anti-rotation device on the valve actuator had failed, and that the valve was not fully closed. This valve is a primary containment isolation valve. An examination of the MOV found that a set screw on the actuator had loosened, allowing the anti-rotation device to slip down the valve stem. When the anti-rotation device slipped far enough, the retainer keys fell out, allowing the valve stem to disengage from the anti-rotation device. The maintenance history of the valve was investigated, and it was found that in 1996, the anti-rotation device was loosened during a scheduled maintenance task. A review of the work documentation package and the vendor manual found that no torque value was specified for the set screw. The Flowserve Vendor was contacted and recommended the set screw be torqued to 60 ft.-lbs. IAW Entergy procedure EN-MA-145, MAINTENANCE STANDARD FOR TORQUE APPLICATIONS. This condition is being reported in accordance with 10 CFR 50.72(b)(3)(v) as a potential loss of safety function of the HPCS system and the primary containment isolation function. An evaluation of the as-found condition has concluded that the HPCS system and primary containment isolation would have been able to perform their design safety function had an actual design basis event occurred during the test.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
River Bend Station – Unit 1	05000-458	YEAR	SEQUENTIAL NUMBER	REV NO.
		2017	004	01

NARRATIVE**REPORTED CONDITION**

On March 23, 2017, at 0028 CDT, with the plant operating at 100 percent power, the high pressure core spray system (HPCS) [BG] was declared inoperable due to a malfunction of a motor-operated valve (MOV) in the system. During a scheduled test, the HPCS pump test return valve (**TV**) to the suppression pool was given a "close" signal after having been opened for the test. The valve position lights indicated that it fully closed, but system flow parameters did not respond as expected. An operator went to the valve and reported that it appeared that the anti-rotation device on the valve actuator had failed, and that the valve was not fully closed. This valve is a primary containment isolation valve and has a design safety function to close automatically upon receipt of a system actuation signal.

This condition is being reported in accordance with 10 CFR 50.73(a)(2)(v)(B) and 10 CFR 50.73(a)(2)(v)(C) as a potential loss of safety function of the HPCS system and the primary containment isolation function.

INVESTIGATION

Operators diagnosing the condition of the valve determined that the anti-rotation device had become dislocated. An examination of the MOV found that a set screw on the actuator had loosened, allowing the anti-rotation device to slip down the valve stem. When the anti-rotation device slipped far enough, the retainer keys fell out, allowing the valve stem to disengage from the anti-rotation device. The valve limit switches were also displaced by this action, resulting in an erroneous indication in the main control room that the valve was fully closed.

In terminating the test, the MOV was given a "close" signal for at least five seconds in accordance with operations administrative guidelines. At this time, system flow parameter returned to normal, and the operator at the MOV reported that it was fully closed. This MOV is a throttle valve, and the extra amount of time under a "close" signal resulted in the valve actually going fully closed.

The maintenance history of the valve was investigated, and it was found that in 1996, the anti-rotation device was loosened during a scheduled maintenance task. A review of the work documentation package found that no torque value was specified for the set screw, whereas the Flowserve Vendor recommends the torque values in EN-MA-145 which requires the set screw to be torqued to 60 ft.-lbs.

An extent-of-condition was performed, and two other similar valves in the HPCS system test return line are susceptible to the same failure mechanism. During work in 2014, the anti-rotation devices for those valves were loosened. The applicable work packages did not incorporate the EN-MA-145 specification to torque the set screw to 60 ft.-lbs. The work instructions only directed that the set screw should be "snug tight."

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CONTINUATION SHEET**

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River Bend Station – Unit 1	05000-458	YEAR 2017	SEQUENTIAL NUMBER 004	REV NO. 01

PREVIOUS OCCURRENCE EVALUATION

River Bend has reported no events caused by a similar failure mechanism in the last three years.

CORRECTIVE ACTIONS TO PREVENT RECURRENCE

The test return valve was repaired, but the anti-rotation device was identified as slightly bent. A compensatory action was put in place for operators to observe valve motion locally until the anti-rotation device is replaced.

An Operability Evaluation has since been performed for E22-MOVF023. RBS-CR-2017-02790 CA7 states "The failure mechanism of this condition was the loosening of the set screw, which ultimately allowed for the anti-rotation device to slide down. Once the anti-rotation device had slid further down, the keys fell out dislodging the shaft from the anti-rotation device. The condition was troubleshot per WO 00470632 which replaced the keys and reinstalled the anti-rotation device in the correct position/location on the stem. The set-screw was properly torqued to 60+ ft.-lbs." The valve is now considered Operable.

A maintenance Action Request (AR 00276966) was initiated and completed to incorporate into future work packages, the vendor recommended torque value IAW EN-MA-145 for installation of the set screw.

Work Orders (WO 00476316 and WO 00476317) to verify the set screw torque on the other two susceptible valves for proper installation have been completed.

SAFETY SIGNIFICANCE

An evaluation of the as-found condition has concluded that the HPCS system and primary containment isolation would have been able to perform their design safety function had an actual design basis event occurred during the test. Upon an automatic system actuation signal, the automatic close signal locks in until the MOV torque switches actuate rather than relying on the limit switches. Even though the anti-rotation device was dislodged, the test return valve would have closed satisfactorily due to the stem packing friction. This event was thus of minimal significance with respect to the health and safety of the public.

(NOTE: Energy Industry Identification System component function identifier and system name of each component or system referred to in the LER are annotated as (**XX**) and [XX], respectively.)